

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

Product Name

Ethylbenzene

Recommended Use  
Uses advised against

Laboratory chemicals.  
No Information available

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | L05908                                                                                      |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                  |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Aspiration Toxicity  
Acute Inhalation Toxicity - Vapors  
Serious Eye Damage/Eye Irritation  
Germ Cell Mutagenicity  
Carcinogenicity  
Reproductive Toxicity  
Specific target organ toxicity - (repeated exposure)

Category 1  
Category 4  
Category 2  
Category 1B  
Category 1A  
Category 1  
Category 2

#### Environmental hazards

Chronic aquatic toxicity

Category 3

### Label Elements



Signal Word

Danger

**Hazard Statements**

H225 - Highly flammable liquid and vapor  
 H304 - May be fatal if swallowed and enters airways  
 H332 - Harmful if inhaled  
 H340 - May cause genetic defects  
 H350 - May cause cancer  
 H373 - May cause damage to organs through prolonged or repeated exposure  
 H412 - Harmful to aquatic life with long lasting effects  
 H319 - Causes serious eye irritation  
 H360 - May damage fertility or the unborn child

**Precautionary Statements****Prevention**

P201 - Obtain special instructions before use  
 P202 - Do not handle until all safety precautions have been read and understood  
 P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
 P233 - Keep container tightly closed  
 P240 - Ground and bond container and receiving equipment  
 P242 - Use non-sparking tools  
 P243 - Take action to prevent static discharges  
 P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
 P271 - Use only outdoors or in a well-ventilated area  
 P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
 P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
 P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
 P308 + P313 - IF exposed or concerned: Get medical advice/attention  
 P331 - Do NOT induce vomiting  
 P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxicity to Soil Dwelling Organisms

## Section 3 - Composition and Information on Ingredients

| Component    | CAS No   | Weight % |
|--------------|----------|----------|
| Ethylbenzene | 100-41-4 | 99.88    |
| Benzene      | 71-43-2  | 0.12     |

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                     |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                              |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                          |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).                                   |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                   |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward. |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                    |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. . Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                                                  |
| <b>Notes to Physician</b>                  | Treat symptomatically.                                                                                                                                                                              |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Hazardous Combustion Products

Carbon oxides.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

#### Environmental Precautions

Do not flush into surface water or sanitary sewer system.

#### Methods for Containment and Clean Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Incompatible Materials**

None known.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

| Component    | New Zealand WEL                                                                                 | Australia                                                                                  | ACGIH TLV                             | The United Kingdom                                                                                                                |
|--------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Ethylbenzene | TWA: 20 ppm<br>TWA: 88 mg/m <sup>3</sup><br>STEL: 40 ppm<br>STEL: 176 mg/m <sup>3</sup><br>Skin | STEL: 125 ppm<br>STEL: 543 mg/m <sup>3</sup><br>TWA: 100 ppm<br>TWA: 434 mg/m <sup>3</sup> | TWA: 20 ppm                           | STEL: 125 ppm 15 min<br>STEL: 552 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 441 mg/m <sup>3</sup> 8 hr<br>Skin        |
| Benzene      | TWA: 0.05 ppm<br>TWA: 0.16 mg/m <sup>3</sup><br>Skin                                            | TWA: 1 ppm<br>TWA: 3.2 mg/m <sup>3</sup>                                                   | TWA: 0.5 ppm<br>STEL: 2.5 ppm<br>Skin | STEL: 3 ppm 15 min<br>STEL: 9.75 mg/m <sup>3</sup> 15 min<br>TWA: 1 ppm 8 hr<br>TWA: 3.25 mg/m <sup>3</sup> 8 hr<br>Carc.<br>Skin |

**Biological limit values**

**NZ** - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**ACGIH** - American Conference of Governmental Industrial Hygienists (ACGIH) TLVs® and BEIs®- Threshold Limit Values for

Chemical Substances and Physical Agents &amp; Biological Exposure Indices. 2022 Edition

| Component    | New Zealand                                                                                                   | Australia | ACGIH - Biological Exposure Indices                                                                                                                                                               | United Kingdom |
|--------------|---------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Ethylbenzene | 0.25 g/g creatinine (urine) end of shift or end of work week (sum of Mandelic acid and Phenylglyoxylic acids) |           | 0.15 g/g creatinine<br>Medium: urine<br>Time: end of shift<br>Determinant: Sum of mandelic acid and phenylglyoxylic acid                                                                          |                |
| Benzene      | 2 µg/g creatinine (urine) end of shift (S-Phenylmercapturic acid)                                             |           | 25 µg/g creatinine<br>Medium: urine<br>Time: end of shift<br>Determinant: S-Phenylmercapturic acid<br>500 µg/g creatinine<br>Medium: urine<br>Time: end of shift<br>Determinant: t,t-Muconic acid |                |

**Appropriate engineering controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment**

**Eye Protection** Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material                                 | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|------------------------------------------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Neoprene, Natural rubber, PVC. | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system.

**Section 9 - Physical and Chemical Properties**

Information on basic physical and chemical properties

|                                                |                               |                                          |
|------------------------------------------------|-------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                        |                                          |
| <b>Appearance</b>                              | Colorless                     |                                          |
| <b>Odor</b>                                    | aromatic                      |                                          |
| <b>Odor Threshold</b>                          | No data available             |                                          |
| <b>pH</b>                                      | Not applicable                |                                          |
| <b>Melting Point/Range</b>                     | -95 °C / -139 °F              |                                          |
| <b>Softening Point</b>                         | No data available             |                                          |
| <b>Boiling Point/Range</b>                     | 135 - 136 °C / 275 - 276.8 °F |                                          |
| <b>Flammability (liquid)</b>                   | Highly flammable              | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                | Liquid                                   |
| <b>Explosion Limits</b>                        | No data available             |                                          |
| <b>Flash Point</b>                             | 15 °C / 59 °F                 | <b>Method</b> - No information available |
| <b>Autoignition Temperature</b>                | No data available             |                                          |
| <b>Decomposition Temperature</b>               | No data available             |                                          |
| <b>Viscosity</b>                               | No data available             |                                          |
| <b>Water Solubility</b>                        | No information available      |                                          |
| <b>Solubility in other solvents</b>            | No information available      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                |                                          |
| Ethylbenzene                                   | 3.6                           |                                          |
| Benzene                                        | 2.13                          |                                          |
| <b>Vapor Pressure</b>                          | No data available             |                                          |
| <b>Density / Specific Gravity</b>              | 0.867                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                | Liquid                                   |
| <b>Vapor Density</b>                           | No data available             | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)       |                                          |

Other information

|                             |                                             |
|-----------------------------|---------------------------------------------|
| <b>Explosive Properties</b> | Vapors may form explosive mixtures with air |
|-----------------------------|---------------------------------------------|

## Section 10 - Stability and Reactivity

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Reactivity</b>                       | None known, based on information available                        |
| <b>Stability</b>                        | Stable under normal conditions.                                   |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |
| <b>Hazardous Polymerization</b>         | No information available.                                         |
| <b>Hazardous Reactions</b>              | None under normal processing.                                     |
| <b>Conditions to Avoid</b>              | Keep away from open flames, hot surfaces and sources of ignition. |
| <b>Incompatible Materials</b>           | None known.                                                       |
| <b>Hazardous Decomposition Products</b> | Carbon oxides.                                                    |

## Section 11 - Toxicological Information

Acute EffectsInformation on likely routes of exposure

**Product Information**

|                   |                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.                                                        |
| <b>Eyes</b>       | Avoid contact with eyes.                                                                  |
| <b>Skin</b>       | Avoid contact with skin.                                                                  |
| <b>Ingestion</b>  | May be harmful if swallowed. Harmful if swallowed. Potential for aspiration if swallowed. |

**Numerical measures of toxicity****(a) acute toxicity;**

|                   |                                                                                |
|-------------------|--------------------------------------------------------------------------------|
| <b>Oral</b>       | Based on available data, the classification criteria are not met               |
| <b>Dermal</b>     | Based on available data, the classification criteria are not met               |
| <b>Inhalation</b> | Category 4<br>Based on available data, the classification criteria are not met |

**Toxicology data for the components**

| Component    | LD50 Oral                | LD50 Dermal                  | LC50 Inhalation               |
|--------------|--------------------------|------------------------------|-------------------------------|
| Ethylbenzene | 3500 mg/kg ( Rat )       | 15400 mg/kg ( Rabbit )       | 17.2 mg/L ( Rat ) 4 h         |
| Benzene      | LD50 = 810 mg/kg ( Rat ) | LD50 > 8200 mg/kg ( Rabbit ) | LC50 = 44.66 mg/L ( Rat ) 4 h |

**(b) skin corrosion/irritation;** Based on available data, the classification criteria are not met

**(c) serious eye damage/irritation;** Based on available data, the classification criteria are not met

**(d) respiratory or skin sensitization;**

|                    |                                                                  |
|--------------------|------------------------------------------------------------------|
| <b>Respiratory</b> | Based on available data, the classification criteria are not met |
| <b>Skin</b>        | Based on available data, the classification criteria are not met |

**(e) germ cell mutagenicity;** Category 1B Based on available data, the classification criteria are not met

**(f) carcinogenicity;**

Category 1A Based on available data, the classification criteria are not met  
The table below indicates whether each agency has listed any ingredient as a carcinogen

| Component    | New Zealand          | Australia             | New South Wales | Western Australia         | IARC     | EU           | UK | Germany |
|--------------|----------------------|-----------------------|-----------------|---------------------------|----------|--------------|----|---------|
| Ethylbenzene |                      |                       |                 |                           | Group 2B |              |    |         |
| Benzene      | Confirmed carcinogen | Notifiable Carcinogen | Notifiable      | Requires Approval for Use | Group 1  | Carc Cat. 1A |    | Cat. 1  |

**(g) reproductive toxicity;** Based on available data, the classification criteria are not met

**(h) STOT-single exposure;** Based on available data, the classification criteria are not met

**(i) STOT-repeated exposure;**

Category 2  
Based on available data, the classification criteria are not met

**Target Organs** No information available.

**(j) aspiration hazard;**

Category 1  
Based on available data, the classification criteria are not met

**Symptoms / effects, both acute and delayed**

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

### Ecotoxicity

#### Aquatic ecotoxicity

The product contains following substances which are hazardous for the environment.  
Contains a substance which is: Toxic to aquatic organisms.

| Component    | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                        | Water Flea                                                                                 | Freshwater Algae                                                                                                                                                                                                                                                | Microtox                                       |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Ethylbenzene | LC50: 7.55 - 11 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 11.0 - 18.0 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 4.2 mg/L, 96h semi-static (Oncorhynchus mykiss)<br>LC50: = 32 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 9.1 - 15.6 mg/L, 96h static (Pimephales promelas)<br>LC50: = 9.6 mg/L, 96h static (Poecilia reticulata)              | EC50: 1.8 - 2.4 mg/L, 48h (Daphnia magna)                                                  | EC50: 2.6 - 11.3 mg/L, 72h static (Pseudokirchneriella subcapitata)<br>EC50: 1.7 - 7.6 mg/L, 96h static (Pseudokirchneriella subcapitata)<br>EC50: > 438 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: = 4.6 mg/L, 72h (Pseudokirchneriella subcapitata) | EC50 = 9.68 mg/L 30 min<br>EC50 = 96 mg/L 24 h |
| Benzene      | LC50: = 5.3 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 10.7 - 14.7 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 70000 - 142000 µg/L, 96h static (Lepomis macrochirus)<br>LC50: = 22.49 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 28.6 mg/L, 96h static (Poecilia reticulata)<br>LC50: 22330 - 41160 µg/L, 96h static (Pimephales promelas) | EC50: = 10 mg/L, 48h (Daphnia magna)<br>EC50: 8.76 - 15.6 mg/L, 48h Static (Daphnia magna) | EC50: = 29 mg/L, 72h (Pseudokirchneriella subcapitata)                                                                                                                                                                                                          |                                                |

#### Terrestrial ecotoxicity

| Component | Earthworm                                                                                                                                                                   | Avian | Honeybees |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| Benzene   | Acute toxicity: LC50 = 0.098 mg/cm <sup>2</sup> (Eisenia foetida, 48 h, filter paper) Acute toxicity: LC50 0.1 - 1 mg/cm <sup>2</sup> (Eisenia foetida, 48 h, filter paper) |       |           |

**Persistence and Degradability** No information available

**Persistence** Persistence is unlikely.

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** Bioaccumulation is unlikely



| Component    | log Pow | Bioconcentration factor (BCF) |
|--------------|---------|-------------------------------|
| Ethylbenzene | 3.6     | 15 dimensionless              |
| Benzene      | 2.13    | 3.5 - 4.4 dimensionless       |

**Mobility** No information available.

#### Other adverse effects

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

#### Waste treatment methods

**Waste from Residues/Unused Products** Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## Section 14 - Transport Information

| Component                          | Hazchem Code |
|------------------------------------|--------------|
| Ethylbenzene<br>100-41-4 ( 99.88 ) | 3YE          |
| Benzene<br>71-43-2 ( 0.12 )        | 3WE          |

#### NZS 5433:2020

**UN-No** UN1175  
**Proper Shipping Name** ETHYLBENZENE  
**Hazard Class** 3  
**Packing Group** II

#### IATA

**UN-No** UN1175  
**Proper Shipping Name** ETHYLBENZENE  
**Hazard Class** 3  
**Packing Group** II

#### IMDG/IMO

**UN-No** UN1175  
**Proper Shipping Name** ETHYLBENZENE  
**Hazard Class** 3

|                                                                                 |                                                                                                                         |
|---------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Packing Group</b>                                                            | II                                                                                                                      |
| <b>Environmental hazards</b>                                                    | No hazards identified                                                                                                   |
| <b>Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code</b> | Not applicable, packaged goods                                                                                          |
| <b>Special Precautions</b>                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| <b>Additional information</b>                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

Any applicable tolerable exposure limits and environmental exposure limits according to the EPA Controls for Hazardous Substances are listed below

| Component | Tolerable Exposure Limit (TEL) Air | Tolerable Exposure Limit (TEL) Water | Tolerable Exposure Limit (TEL) Surface | Environmental Exposure Limits (EEL) |
|-----------|------------------------------------|--------------------------------------|----------------------------------------|-------------------------------------|
| Benzene   | 0.01 mg/m <sup>3</sup>             | 0.01 mg/L                            |                                        | 2000 µg/L (Water)                   |

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

| Component | New Zealand          |
|-----------|----------------------|
| Benzene   | Confirmed carcinogen |

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                                                                                                                                                                                           | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Benzene   | -                                                                   | Use restricted. See item 72. (see link for restriction details)<br>Use restricted. See item 5. (see link for restriction details)<br>Use restricted. See item 28. (see link for restriction details)<br>Use restricted. See item 29. (see link for restriction details)<br>Use restricted. See item 75. | -                                                                                                     |

|  |  |                                    |  |
|--|--|------------------------------------|--|
|  |  | (see link for restriction details) |  |
|--|--|------------------------------------|--|

<https://echa.europa.eu/substances-restricted-under-reach>

### International Inventories

China, X = listed, Australia, U.S.A. (TSCA), Canada (DSL/NDL), Europe (EINECS/ELINCS/NLP), Australia (AICS), Korea (KECL), China (IECSC), Japan (ENCS), Philippines (PICCS), Taiwan (TCSI), Japan (ISHL), New Zealand (NZIoC), Japan (ISHL). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component    | CAS No   | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|--------------|----------|-------|------|-----------|--------|-----|----------|-------|------|
| Ethylbenzene | 100-41-4 | X     | X    | 202-849-4 | -      | -   | KE-13532 | X     | X    |
| Benzene      | 71-43-2  | X     | X    | -         | -      | -   | KE-02150 | X     | X    |

| Component    | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|--------------|----------|------|-----------------------------------------------|-----|-----|-------|------|------|
| Ethylbenzene | 100-41-4 | X    | ACTIVE                                        | X   | -   | X     | X    | X    |
| Benzene      | 71-43-2  | X    | ACTIVE                                        | X   | -   | X     | X    | X    |

**Legend:** X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

### Legend

**NZIoC** - New Zealand Inventory of Chemicals

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**IECSC** - Chinese Inventory of Existing Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**NZS 5433:2020** - Transport of Dangerous Goods on Land

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**WEL** - Workplace Exposure Limit

**DNEL** - Derived No Effect Level

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**CAS** - Chemical Abstracts Service

**ACGIH** - American Conference of Governmental Industrial Hygienists

**PNEC** - Predicted No Effect Concentration

**OECD** - Organisation for Economic Co-operation and Development

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**LC50** - Lethal Concentration 50%

**ATE** - Acute Toxicity Estimate

**RPE** - Respiratory Protective Equipment

**NOEC** - No Observed Effect Concentration

**BCF** - Bioconcentration factor

**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

**Classification and procedure used to derive the classification for mixtures according to Regulation (EC) 1272/2008 [CLP]:**

**Physical hazards** On basis of test data

**Health Hazards** Calculation method

**Environmental hazards** Calculation method

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and

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hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

|                  |                 |
|------------------|-----------------|
| Revision Date    | 22-Mar-2023     |
| Revision Summary | Initial Release |

#### Disclaimer

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## End of Safety Data Sheet